Biagio Scarciello

Embedded Software & Systems Engineer

C/C++ | Real-Time Systems | Embedded Linux | Railway Safety-Critical Systems

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Profile

Senior Embedded Software & Systems Engineer with 8+ years of experience delivering safety-critical railway systems across high-speed, metro, and regional platforms. Expert in Embedded C/C++, real-time system design, and onboard communication protocols (TRDP, IPTCom, Ethernet). Experienced across the full engineering lifecycle: system architecture definition, integration, verification & validation, and on-site commissioning. Complemented by hands-on full-stack development (Python/FastAPI, React, PostgreSQL, Docker) applied to production systems.

Work Experience

Software Lead — Diagnostic Systems 2024 – Present
Hitachi Rail STS S.p.A. *Naples, Italy*

- **Software lead** for the diagnostic systems of the **Rome Metro (lines A/B)** and **Florence Tram** projects, owning the software from development through integration and on-site commissioning.
- **Software development support** on the **Milan Metro (lines 1, 2, 3)** diagnostic system, contributing to development, integration, and issue resolution.
- Led a focused stabilization phase on the Rome Metro project ahead of official customer testing, **resolving 230 open issues within one month** and securing the validation milestones on schedule.
- Led the full software lifecycle of onboard diagnostic systems, developing the HMI and optimizing the embedded backend framework in Embedded C.

System Engineer — Event Recorder & Safety-Critical Subsystems 2022 – 2024
Hitachi Rail STS S.p.A. *Naples, Italy*

- System owner of the **Event Recorder** product line: defined the complete system architecture and requirements, and drove the solution from specification through integration, validation, and customer acceptance for the metro networks of **Milan, Rome, and Washington (WMATA)**.
- System Engineer referent for Bogie Control System communication (TCMS integration) using message data protocol, defining the communication architecture and coordinating cross-functional validation activities across software, hardware, and RAMS teams.
- Technical referent for onboard safety-critical subsystems — Event Recorder, Dead Man, and Zero Speed detection — covering requirements analysis, traceability (DOORS), interface definition, and verification strategy.
- Coordinated integration, troubleshooting, and commissioning of diagnostic subsystems in operational environments, improving the reliability of diagnostic data communication between subsystems and TCMS.

Embedded Software Developer — Regional & High-Speed Platforms Apr. 2018 – 2022
Hitachi Rail STS S.p.A. *Naples, Italy*

- Developed and commissioned the diagnostic systems for the regional fleets **Trenitalia Rock**, **Trenitalia Blues**, and **Ferrovie Nord Milano**, and contributed to the high-speed platforms (*ETR1000 Frecciarossa, ETR700*).
- Developed and extended Condition-Based Maintenance (CBM) software in Embedded C, and designed diagnostic algorithms in C and MATLAB/Simulink for onboard systems.
- Enhanced and refactored the embedded diagnostic software framework, improving maintainability and reuse across train platforms.
- Implemented multi-vehicle composition management logic (backend and frontend) and designed diagnostic HMI components in HTML/JavaScript.
- Improved the reliability and stability of distributed onboard communication networks, managing issue tracking and defect resolution through Jira during deployment phases.
- Supported integration, validation, and troubleshooting activities during commissioning phases in operational environments.

Full-Stack Developer Apr. 2024 – Present
Bispace *Naples, Italy*

- Bispace is a personal side project supporting professionals in navigating digital complexity with a clear, human-centered approach.
- **Liftoff** — web application for a professional coach to help clients follow structured training programs and track performance over time, allowing users to access workout plans, monitor progress, and record results. (Python, Flask, HTML, JavaScript)

- **Multicon** — embedded network management application for OpenWRT-based routers (Teltonika RUT955), enabling remote configuration via MQTT JSON commands, with a Flask-based manager server offering a web interface and orchestrating MQTT rooms (topics). (C, Python, Flask, OpenWRT SDK, Mosquitto, JSON, MQTT, WireGuard VPN, Networking)

- **FoxOffice** — web platform to centralize business operations, integrating a CRM module, document management, process automations, and performance dashboards into a single structured environment. (FastAPI, React)
- **NAPP** — multi-tenant HACCP compliance platform for food businesses, built as a monorepo: FastAPI backend, React owner panel for client/document/equipment configuration, and a Flutter tablet app used by operators to log temperatures, production, sanitation, and non-conformities. (FastAPI, PostgreSQL, Alembic, MinIO, React, TypeScript, Flutter, Docker)

- **CRM Evolution Time** — multi-workspace web platform for watch inventory management, with role-based access control (Owner, Admin, Warehouse, Accountant), JWT refresh-token rotation, Excel bulk import, and integration with the OKTOPIA marketplace. (FastAPI, SQLAlchemy 2 async, PostgreSQL, Next.js 15, TypeScript, React Query, Docker)
- **CRM Financial** — accounting management platform for an accounting firm, handling invoices, purchases, daily takings, and consolidated financial summaries, with document storage and multi-client management. (FastAPI, SQLAlchemy async, PostgreSQL, MinIO, React 19, TypeScript, Vite, Docker)
- **Medify** — web application for gynecologists to manage patient records, visits, and pregnancy monitoring.
- **WorkoutLog** — AI-powered web application for tracking and analyzing workouts; users upload sessions via text or image (screenshot), and the system leverages OpenAI APIs to provide personalized feedback and training suggestions based on the user's history and results.
- **Websites** — Sabe SNC, Swinginnapoli, Dott.ssa Federica Terzi, Jtf Gaskets, DV Barberia, Bpnutrition.

Education

M.Sc. in Automation Engineering
University of Naples Federico II | 110/110 cum laude — *Naples, IT*
B.Sc. in Automation Engineering
University of Naples Federico II | 107/110 — *Naples, IT*
Full Stack Development & AI Agents Specialization — In Progress
Start2Impact University (Online)

Technical Skills

Embedded & Real-Time Systems: Embedded C/C++, real-time systems design, memory management & performance optimization, embedded Linux environment, debugging & troubleshooting in field environments
Communication & Distributed Systems: Ethernet-based onboard networks, TRDP (Train Real-time Data Protocol), IPTCom, MQTT, TCP/IP networking, multi-vehicle communication logic, communication gateway design
System Engineering & Validation: System architecture definition, Integration/Verification & Validation (IV&V), commissioning support, requirements analysis & traceability (DOORS), issue tracking (Jira), safety-critical environment exposure (railway domain)
Programming Languages: C (Embedded), C++, Python, JavaScript/TypeScript, Dart
Web & Full-Stack: FastAPI, Flask, React, Next.js, Flutter, SQLAlchemy, PostgreSQL, REST API design, JWT authentication & RBAC, Docker & Docker Compose
Development Tools: CMake, Git, SVN, GitHub, Wireshark, CI/CD fundamentals, MATLAB/Simulink

Languages: *Italian*, *English*
Interests: Crossfit, Swing, Technology, Graphic, Travels

Publications

Model-based vehicular prognostics framework using Big Data architecture, *Computers in Industry, Elsevier*, A. Petrillo, A. Picariello, S. Santini, **B. Scarciello**, G. Sperlì, (2019) - doi.org/10.1016/j.compind.2019.103177

- Proposed **ATOMIC** (Automotive predicTOr Maintenance In Cloud), a cloud-based Big Data infrastructure for Integrated Vehicle Health Management (IVHM) that combines nonlinear discrete-time models of nominal and faulty component behaviour to estimate End-Of-Life (EOL) and Remaining Useful Life (RUL) indicators on-line.
- Validated the prognostic method on the Delphi DFG1596 fuel pump and benchmarked the architecture's deployment configurations in terms of scalability and cost.